



Master Thesis Manual

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1 Introduction

This document serves as the course manual for students who will complete their master graduation project in Computational Science. The guide is primarily intended for the use of students. However, it also explains the roles of the graduation supervisor, second assessor, and examiner. The master thesis project is a mandatory part of the Computational Science master program (42 EC). The student is expected to carry out independent research at a senior level. In this project, the student will work on a predefined set of project goals and must create the required products within the stated period. The thesis project begins with students finding a suitable topic, a daily supervisor, and an examiner (eventually a second reviewer is also required – see section 3 on roles). Students should begin working on their master thesis part-time in November-December of the second year. This is the period in which they should conduct a literature study. This means students should have confirmed the topic, daily supervisor, and examiner by October at the latest. A detailed overview of planning activities is given in section 2. The master thesis project can be done within the Computational Science research group, another research centre at UvA, or at an external research institute. For projects external to the UvA, students are also required to find an examiner who is one of the appointed UvA staff examiners (see here for the official list of examiners). In order to help students find a supervisor and/or examiner, information about lecturers and their field of expertise can be found here, or browse the study guide to check all courses per examiner. Most important to keep in mind when a student starts the master thesis project: (1) a suitable arrangement has to be made with the daily supervisor and examiner, (2) a timeline should be set up and (3) the student

and supervisor must write a brief project description. All of these items are to be submitted on DataNose by the student prior to starting the master thesis.

1.1 Objectives

In completing the MSc Computational Science thesis, you will develop the following academic and research skills:

- The student formulates a research question.
- The student studies relevant literature.
- The student analyses research results using skills and techniques acquired during the master's programme and shows that they can connect theories, models, and concepts with and within the research topic.
- The student answers the research question using analytical and numerical methods.
- The student is capable of independent and critical reflection on their own and on other relevant work.
- The student can write a report that clearly describes the research question, motivation for this research, review of relevant literature, methodology used, results, conclusions, and recommendations.
- The student can give a clear presentation about the research question, motivation for this research, methodology used, results, conclusions, and recommendations.
- The student can collect data for validating and/or fitting the model and implement and verify computer programs.

2 Project Timeline

2.1 Important dates

The master thesis is a 42 EC course, which means 7 months of full-time work. More typically this will run as 2 months of part-time (literature study Nov-Dec) work and 6 months of full-time work (thesis work Jan-Jun). The exact timelines may differ, but all students should aim to work along the normal schedule. This will ensure completing the programme within 2 years and prevent the necessity for further tuition payment. Also, keep in mind that not all examiners will be available in the summer to join your thesis defence if you plan to delay your thesis. There are multiple things to consider in your thesis timeline.

- Four months before start: Read all the information about the Master Project. Ask the coordinator any questions regarding the Master Project. Start thinking about your thesis subject and whether you want to conduct your thesis within the university or an external company.

- Two months before start: At least one month before assuming starting the thesis project: request approval by submitting the proposal on DataNose.
- Before starting: The student must have received approval via DataNose to start the project.
- The first month: Start literature research and set up an action plan or timeline for yourself (which does not have to be submitted).

3 Roles

During your thesis project, you are required to have a daily supervisor, an examiner, and an assessor (second reviewer). These are roles which can be assigned to members of staff or researchers at the institute or company where the project is performed. Before starting your thesis, it should be clear who your examiner and (daily) supervisor are. It is possible that the second assessor will be appointed later in the project. Eventually, at your thesis defence, your thesis committee should consist of three people at minimum. This will include your examiner, daily supervisor and second assessor. Please note that the roles of examiner, daily supervisor, and second assessor should be fulfilled by three different people. No two roles can be fulfilled by the same person.

3.1 Student

What is expected of the student? An active attitude and independence: the student is the process ‘manager’ of the project; it is not the task of the supervisor to chase the student, let alone help the student achieve a pass. So, if you show no or too little initiative, you will get an unsatisfactory result.

- Initiative: the student makes regular appointments with the supervisor (at least once every two weeks); the student contributes to the progress.
- Active planning: the student works regularly and in a structured way on the project; the student adjusts the schedule of the project if necessary; the student ensures that she/he comes prepared to the appointments with the supervisor. Proper preparation with talking points, specific questions, or visual aids greatly increases the productivity of these meetings.
- Communication - talking to others: It helps to talk to others about your project, for example, to explain where and why you are stuck. The other person may not understand much of it, but the process of describing or explaining will help you understand areas that require more work/research. Fellow students are excellent at this, but family and friends are also good.
- Supervisor management: each supervisor has strengths, but they often have multiple teaching tasks, research projects, and other obligations. Please be mindful and respectful of this.

- Punctuality - Start writing in time: a thesis is rarely written in one go; it is a process of trial and error. It is advisable to start writing early, for example by keeping a log. It is tempting to think that you will remember all your hunches and lines of reasoning, but you will be grateful at the end of your project that you kept thorough notes, especially if the project evolved in an unanticipated direction. A well-written thesis almost always requires a greater portion of time than expected.

The student should lead the project. This means that they are responsible for arranging meetings with the supervisor and examiner, and if necessary, the assessor. The students are also responsible for sending their thesis in time for feedback to the daily supervisor. They should submit a thesis in DataNose at least two weeks prior to their defence for final evaluation. In cases where the daily supervisor is external, for example, in a company or another university, the student is responsible for communicating the role of the daily supervisor and the expectations. It is encouraged that students send the link for the thesis manual to their daily supervisor, so that the supervisor may become well acquainted with the role and surrounding procedures.

3.2 Daily supervisor (external or internal)

Acting as the daily supervisor of a thesis project involves day-to-day supervision and scientific guidance of the project. There are no official restrictions on this person, except the condition that the supervisor should be someone with sufficient expertise (on the topic of your thesis project) to perform the scientific guidance. The daily supervisor can be:

- a PhD student or a postdoctoral fellow at the Faculty of Science (FNWI),
- a researcher at other institutes/universities, or
- an employee at the company where the thesis project is performed.

The supervisor plays a central role in organising the supervision of the master's thesis. The supervisor provides an intellectual environment in which the master's student can develop his or her research. The primary job of the supervisor is to give feedback regarding the research question, the methodology, and the results in line with the expected outcome of a CLS thesis. The essence of giving feedback is that the supervisor does not propose solutions but identifies problems. Students are expected to take the initiative to make sense of the feedback and propose their own solutions to the problems. If for any reason the student is not satisfied with the supervision, the student should contact the thesis coordinator and the examiner directly. No general rules on the frequency of supervision meetings exist – instead, the frequency depends on the individual agreements between the student and the supervisor. It is the student's responsibility to signal the need for meetings with the supervisor in a timely fashion. Most students meet their daily supervisors once every week. The daily supervisor will attend the thesis defence and participate in grading the thesis

as outlined in sections 9 and 10. **It is the responsibility of the student to find a suitable thesis supervisor.**

3.3 Examiner (internal)

The role of the examiner has one important restriction: the role should be fulfilled by an official examiner listed on the official List of Examiners by the Examinations Board of MSc Computational Science. The examiner of a thesis project will determine the final grade of the thesis project. The List of Examiners can be found in DataNose and Canvas.

- Ensure compliance with the learning outcomes of CLS, graduate school policies and regulations, including the key policies and procedures relating to examination.
- To grade and write a brief report on the candidate's thesis. The report should be written after reading the thesis and attending the defence. This will be written in consultation with the defence committee as outlined in sections 9 and 10.
- To give the candidate adequate opportunity, encouragement, and time to explain the research and to defend the thesis during the thesis defence.

It is the responsibility of the student to find a suitable thesis examiner.

3.4 Assessor (external or internal)

The second assessor should be someone with sufficient expertise regarding the topic of the thesis project. This role does not involve any supervision. Often the daily supervisor or the examiner of a thesis project can help find (make suggestions about) a suitable second assessor for a thesis project. Please note: the role of the examiner and the role of the second assessor for a thesis project cannot be fulfilled by the same person.

3.5 Thesis Coordinator (TC)

The Thesis Coordinator (TC) is responsible for the overall organisation of the MSc Computational Science thesis program. The TC monitors and approves all thesis project proposals – both the projects listed on the marketplace as well as those submitted by students via DataNose. The role of the TC also involves answering general questions about the thesis project that cannot be found in this guide or online and provides clarification when required. The TC will provide feedback and support to supervisors and examiners on supervision and evaluation.

3.6 Study Advisor

The role of the Study Advisor is to answer questions students may have on effective studying as well as help students if they run into challenges due to personal circumstances. The Study Advisor will be able to help you create a personal plan to help your studies run smoothly. Extensions can be proposed by the Study Advisor based on personal circumstances. If needed, the Study Advisor will also mediate with students and supervisors.

4 Before starting your thesis

Before starting your master thesis project, you need to obtain permission from the Examinations Board, and you need to submit your thesis topic (research proposal) on DataNose using the link <https://datanose.nl/yourprojects>. The template for the research proposal is available on the Canvas course page.

4.1 Obtaining your permission from Examinations Board

To start your master thesis project, you are required to have completed all core and constrained choice courses of the programme. Furthermore, you need to have a plan for the rest of your (elective) courses and your daily supervisor and the examiner should agree with your course list. You need to fill in the Study Plan Application (SPA), which can be found in DataNose on your personal page under the heading PROGRESS (NL: voortgang) or login at <https://acadplan.datanose.nl/>. Please note: you will not be granted a diploma if you have not asked the Examinations Board for formal approval of your personal curriculum.

4.2 Study Plan Application (SPA)

You can use the SPA to:

- Check your study progress
- Indicate your study plan
- Send a request to the Examinations Board concerning your personal study programme
- ask the examinations Board for approval of your personal curriculum in order to graduate.

When you are logged in at <https://acadplan.datanose.nl/> you can:

- See which courses you have already passed
- Indicate which courses you still intend to complete
- Tick the courses you intend to complete to include them in the study plan

- Submit a request to replace a course or to request an exemption by means of the arrow on the right side next to the course. This request will automatically be sent to the Examinations Board.

If you have questions about the SPA please send an email to:
master-support@computationalscience.nl

4.3 Requirements and Exemptions

- Again, the student may start with the final project of the study programme (master thesis project) only if all other compulsory components have been completed and the student has completed all necessary restricted-choice courses (30 EC). The student's final study programme must also have been approved by the Examinations Board.
- In case one or more courses are still to be completed, the supervisor and the examiner may agree that the student may start with the master thesis project (by approving the project proposal submitted by the student in DataNose)

5 How do I get a graduation project topic?

5.1 Topics

A research topic must be related to Computational Science. Examples of fields of topics are:

- Modelling and simulation of real life applications
- Analysis of complex systems and phenomena
- Computational experiments
- Data-driven modelling
- Advanced numerical algorithms
- Parallel and distributed computing
- Virtual laboratory, advanced visualization

Note: a lot of subjects also include machine learning. However, a topic that is an only machine learning based is not sufficient. The same applies to pure software engineering, pure deep learning or AI, NLP, computer vision, and information systems. You may suggest a topic extension that does build on these subjects using computational science.

5.2 Examples of past papers

Examples of MSc CLS thesis project topics are:

- Understanding biomass dynamics in semi-arid ecosystems: A computational means to an organic end.
- Causal Discovery from Spontaneous Targeted Interventions
- Effects of internal viscosity contrasts in a blood flow model based on immersed boundary lattice Boltzmann methods
- Optimizing Resource Allocation in Socio-economic Systems with the Minority Game: A Case Study on Electric Cars

You can find publications of previous students at [this link](#).

5.3 Thesis Fair

The Thesis Fair is an opportunity for you to meet with organisations and discuss their offered projects with the hope that you will find your thesis project. To participate in the Fair, you must vote for organisations that you wish to meet on a web platform. The organisations include local and international companies, start-ups, government agencies, and non-profits. You vote on individual projects and then the organisation appears on your voting list. A schedule will be generated for you based on your ranked top 15 projects. Information Session: Ahead of the Thesis Fair, there will be an Information Session webinar. This session will inform students about the expectations for the upcoming Fair and for the thesis itself. A presentation will be given on what to expect, how to vote for projects, and how to prepare. Thesis Marketplace (voting): A web-based Thesis Marketplace provides you with available projects. Students will be able to vote on the same page as the Marketplace. When arranging the speed dating schedule, the aim is to match students with the companies and projects coinciding with their votes. Unfortunately, securing a speed date with every company voted for is impossible. Please note that there is a chance you will not be able to meet with some of your top organisations if they are overly popular. For more information about the Thesis Fair can be found on the Thesis Fair website, where you will find all the important information. Please read through the FAQ sections that you can find under the ‘Students’ section for details on any questions that you might have. If you have any additional questions that are not listed on the website please contact the Thesis Fair team at thesisfair-IvI@uva.nl.

5.4 Submitting your own project proposal

Students can submit their own project proposals. Students should prepare a very clear research proposal (see the template on Canvas) and find an examiner, daily supervisor, and assessor for the thesis project. Once agreed upon, students

should submit the research proposal in DataNose before the end of Block 2 Year 2.

6 Thesis procedures and processes

6.1 Approval of topic

Once you have completed the form on DataNose an e-mail will go to the thesis coordinator as well as the examiner. They both must approve your proposed topic and agree with their roles appointed by you. **This step is mandatory before commencing your thesis research.** Completing this form is important for all students as the thesis grading will be done via DataNose.

6.2 Thesis or Internship Agreements

For projects that are completed at an external organisation, you may need to complete a Thesis or Internship Agreement. For this, you will need an authorised signature from the UvA. If a Thesis Agreement is required, please read the details here: <https://student.uva.nl/en/topics/internship-possibilities-and-procedure> The procedure, in short—the programme director signs the standard model contract and any changes or addendum are escalated to Legal Affairs and the Director of Operations through the email address: internshipagreement-science@uva.nl

7 The Thesis Structure/Layout

7.1 Template

The master thesis report should be written using the template available in Overleaf. Include the chapters of the thesis as separate .tex files from the Chapters folder. Change the file names if you prefer. This structure provides the bare-bones essentials of the thesis and some indicative length. This structure may not fit your thesis perfectly but be sure to include all the components in this template somehow. It is possible to split the chapters up (E.g., 2-methods chapter, Experiment and Results as separate chapter, etc.). The typical length may be between 50 - 70 pages. Do not worry if the end result is slightly larger or smaller than this, but a thesis of 20 pages or 100 pages may suggest that you have been too brief or too verbose. Note that there is no strict limit/minimum.

7.2 Structure

- The thesis will need to include the following sections:
- Introduction: Set out your thesis and state your research question (5-8 pages)

- Literature Review: Summarize the base of related literature and material (8-12 pages)
- Methods: Provide details of your approach (10-15 pages)
- Experiments and Results: Communicate the outcome of testing your hypothesis(es) (10-15 pages)
- Discussion: Evaluation/analysis of the meaning of findings and how they fit in the greater context of the field; outline limitations(5-8 pages)
- Conclusions and Future Work: 5-6 pages

8 Thesis submission procedure

8.1 Submitting your thesis on DataNose

Students are expected to upload their final thesis in DataNose at least two weeks prior to their defence. A hard copy of the thesis is not required.

8.2 Late submission penalties

Students are expected to finish their master thesis research project within the ‘normal’ time limit of 7 months plus a 2-month grace period (a total of 9 months). The final grade of a student who takes longer than these 9 months to finish his or her project will be subjected to a penalty. Any student who takes longer than 9 months receives a half-point deduction for every extra month taken, up to a maximum of 2 points. This means, for example, a student who receives an 8.5 in month 11, would have their grade reduced by 2 times 0.5, yielding 7.5. A student who receives a 9 in month 15, would receive a half-point deduction for each extra month (but to a maximum of 4 months) resulting in a 7. **The penalty is enforced by the examiner in consultation with the daily supervisor.**

8.3 Extensions and suspensions of the thesis

There are circumstances where students will need to suspend their thesis or will need an extension. Students are allowed to (formally) ask for an extension or suspension of their master thesis research project via DataNose. Please keep the study advisor, daily supervisor, and examiner in the loop. Extension or suspension of a master thesis project is only granted to those with strong reasons. These requests (and motivations) must be approved by the examiner and the student in question must have discussed the reasons with the study advisor before asking for an extension. Students should also inform their daily supervisor and examiner about the delay and the request for an extension. In cases where planned suspension is not possible (e.g., unresolved personal/health issues) it is advisable to contact the study advisor. In cases where the requests

are personal, the study advisor will collate advice from all parties and make the final decision. The study advisor may also ask the opinion of the Programme Director and the Examinations Board where necessary. In most cases, if both the daily supervisor and examiner approve the suspension, the student will be permitted to delay. The extension request must be discussed and submitted as soon as the relevant issue arises, not at a later date. For students with an extension, the grading procedure is expected to take longer due to the timing of teaching periods.

9 Defence

The defence of your Master Thesis can be organised when the examiner and daily supervisor indicate that the work and the report are to an acceptable standard. The final assessment takes place after the report (thesis) has been submitted and the project has been presented and defended. The following matters should be arranged a month before the expected graduation date:

9.1 Appointing a defence committee

You and your daily supervisor/examiner should identify a defence committee to assess your thesis. A committee must comprise of a minimum of three people. This will include your examiner, daily supervisor, and second assessor.

9.2 The defence

Once you have finished your research and have uploaded the written thesis report (which is approved by your examiner and daily supervisor) in DataNose, you are ready to defend your Master Thesis Project! Please make sure the thesis report is uploaded in DataNose and available to the Committee members two weeks (14 days) before the Master Thesis defence. Students can book their rooms themselves. Many rooms B*.* are available to students via <https://uva.mapiq.net/>. If these are insufficient, then students can ask ESC or the programme coordinator for a solution.

9.3 Presentation

The examination session contains 3 parts and will take approximately 1.5 hours from start to finish:

- Your presentation about your work (in English) should take approximately 25 minutes. This presentation should strive for a nice balance between a general introduction (so that the lay members of the public know what you have done and why) and a more technical part for your peers. It is not necessary to include every specific detail, because the committee members have read your thesis. After the presentation, the audience can ask you questions.

- Questions will be accepted from the Defense Committee for about 30 minutes. Each member will ask 2 or 3 questions on your thesis and work, moderated by the session chair. Experts typically ask detailed questions, whereas the 'outside' member may want you to take a more contextual view of what you have done.
- The committee will then consult in private for 30-45 minutes to decide your grade using an evaluation form with supporting information from your daily supervisor. When this concludes, you will be invited back into the room, and they will explain the final grade to you (see section 10 for more details). Upon granting a positive result, your daily supervisor is invited to speak some congratulatory words.

10 Assessment

Once the defence presentation has been completed the grade is determined by the committee members and submitted in DataNose.

10.1 Assessment criteria

The grading will be based on the rubric as provided in Canvas. You can also access the rubric [here](#).

10.2 Execution of the assessment

The examiner of the thesis will have access to enter the grades in DataNose. As shown above the assessment has three broad components. The first part is your daily research work which constitutes 35% of the grade. The examiner will primarily consult the daily supervisor to grade this part which can be done before or after the thesis defence presentation. The second part is your written report which constitutes 50% of the grade. The examiner will consult the daily supervisor and the assessor and use examiners own judgement to grade this part which can be conducted after everyone has read the thesis submitted in DataNose or after the thesis defence presentation. The final part is your presentation, which constitutes 15% of the grade. The examiner will consult the daily supervisor and the assessor and use her or his own judgement to grade this portion, which can be done only after the thesis defence presentation. Usually, the committee takes about 30 – 45 minutes to deliberate on the grade after the defence presentation.

11 Important Information

11.1 Contact Information

- mscclsthesis-gsi@uva.nl for any thesis related questions.

- Study Advisor
- servicedesk-esc-science@uva.nl for general questions about studying at the UvA
- Examinations Board
- Student Service Desk

11.2 Links

- MSc Computational Science Thesis website
- List of Examiners
- Internship Agreement
- Graduating and degree certificate